

JAW REFLEXES & REFLEX CONTROL OF MASTICATION

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LECTURER

DIKIOHS, DUHS.

LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- Define mastication & the muscles involved in mastication.
- Identify & name different phases of mastication.
- Describe the role of trigeminal nerve in sensory & motor control.
- Define reflex & types of jaw reflexes.
- Identify the location & function of masticatory central pattern generator(CPG) in mastication.
- Describe how reflexes can continue chewing automatically once initiated voluntarily.
- Relate mastication dysfunction to chewing difficulty.

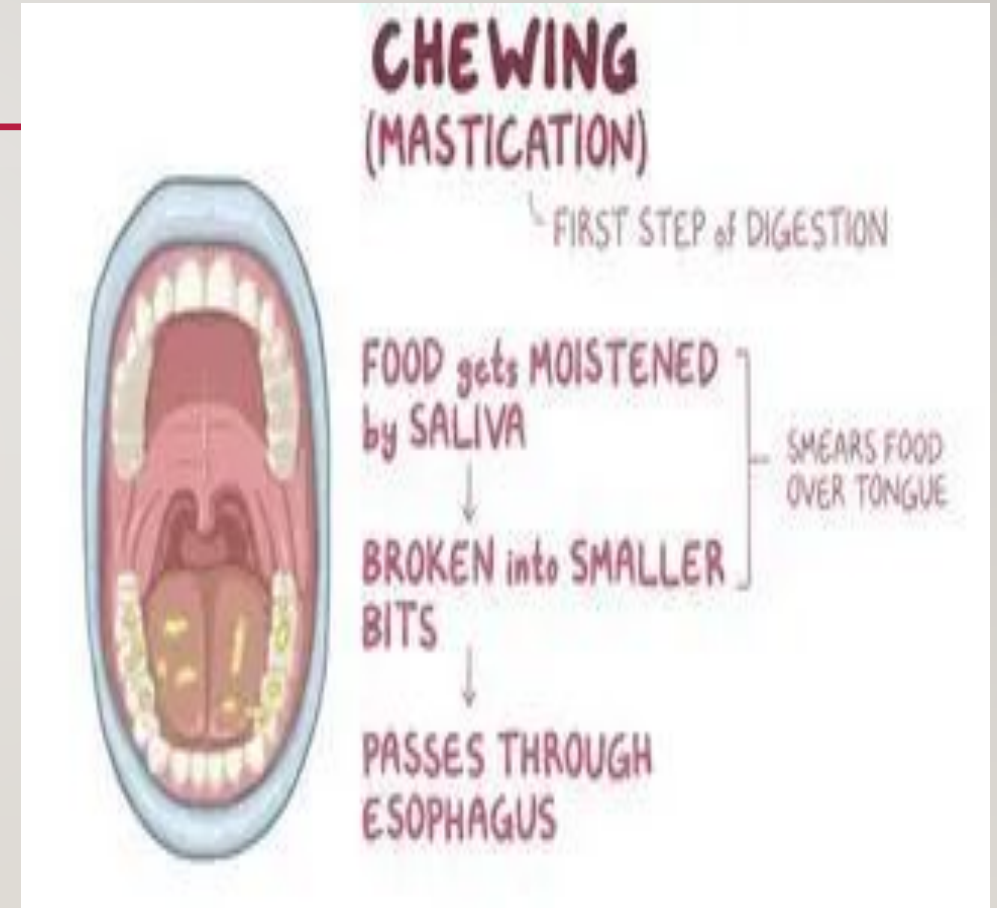
WHAT IS MASTICATION?

It is a neuromuscular activity involving vertical & horizontal movements of the jaw, teeth, tongue & cheeks to crush food into manageable, moist particles to prevent choking & aids digestion.



INTRODUCTION

- ❑ Mastication is the initial stage of digestion.
- ❑ During mastication, food is mechanically broken down in the mouth by the action of teeth, tongue & jaw muscles.
- ❑ It breaks down food into smaller particles to increase the surface area for the action of enzymes & mixes it with saliva to form bolus.
- ❑ Mastication facilitates swallowing & prepare the food for stomach.



□ Mastication is a complex neuromuscular process initiated voluntarily followed by reflex control.

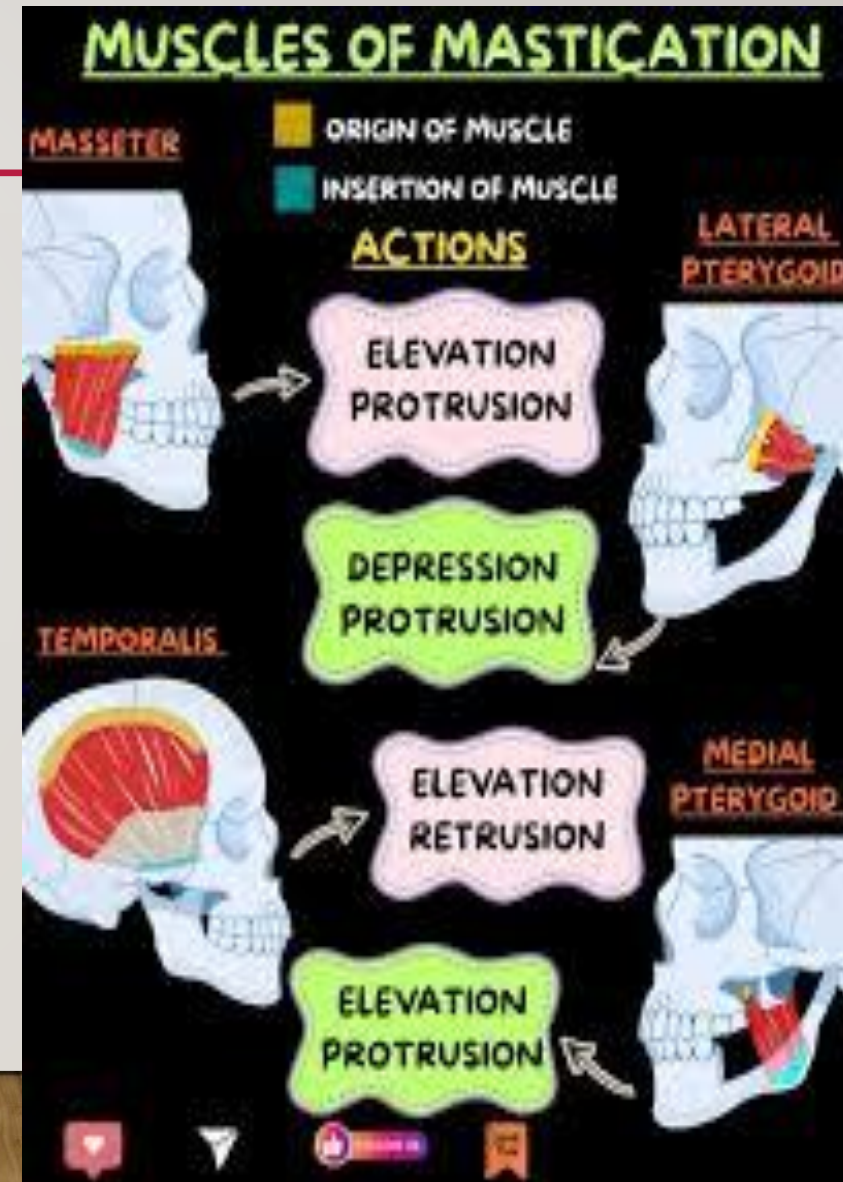
□ Mastication is not a simple process of just chewing, it requires:

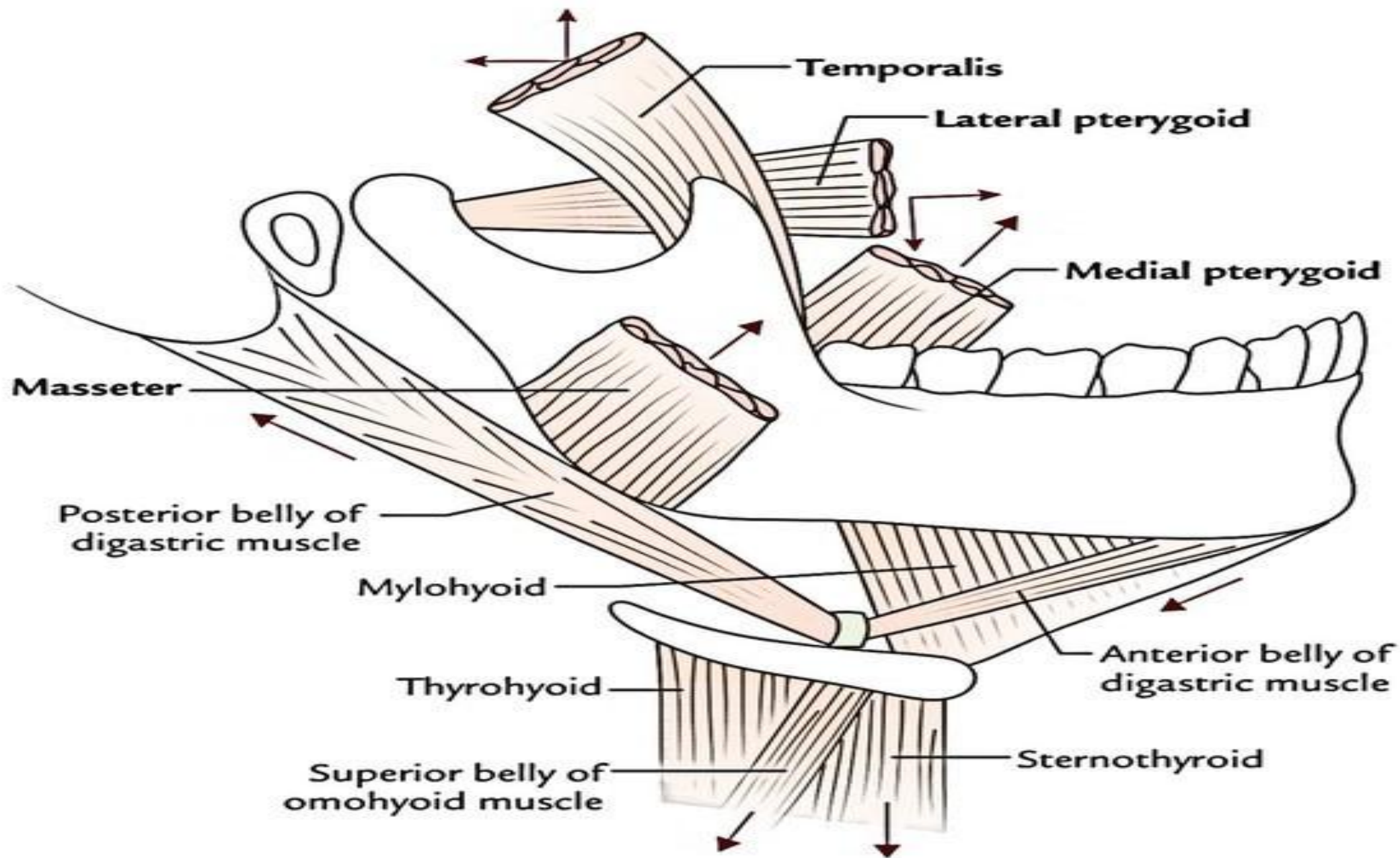
1. Co-ordination of muscles
2. Sensory feed-back
3. Brain control

□ Mastication is initiated voluntarily but automatic adjustments during chewing is a reflex controlled activity e.g. when we chew something hard, our body automatically adjusts the force.

MUSCLES INVOLVED IN MASTICATION

- There are four muscles involved primarily in the process of mastication hence named ***muscles of mastication***.
- ***Masseter***: it is the main muscle for jaw closure & generate strong forces.
- ***Temporalis***: it is involved in the elevation & retraction of the jaw.
- ***Medial pterygoid***: this muscle works with masseter muscle to cause elevation of the jaw.
- ***Lateral pterygoid***: it opens the jaw & also produces side to side movements.
- All muscles of mastication are innervated by the mandibular branch of trigeminal nerve.





PHASES OF MASTICATION

□ The chewing process(mastication) has three main phases:

1. Opening phase
2. Closing phase
3. Occlusal (intercuspal) phase

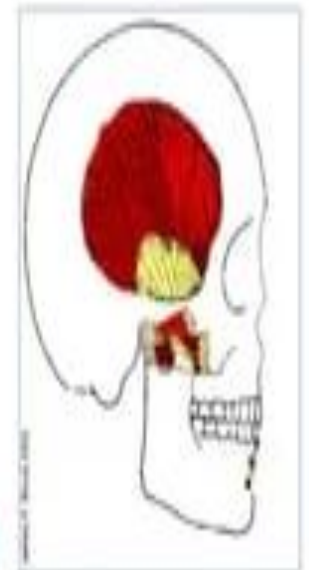
OPENING PHASE: Opens the mouth by lowering of the jaw & food enters between teeth.

CLOSING PHASE: Jaw rises i.e the mouth is closed & food is positioned between the teeth.

OCCLUSAL PHASE (INTERCUSPAL): During this phase, grinding of food takes place by the contact of teeth with the food.

MASTICATORY CYCLE

- Pathway of mandible in chewing.
- Opening phase- teeth & condyle downward & forward.
- Closing phase- elevated, lateral movement of mandible.
- Intercuspal phase- forward and medial.

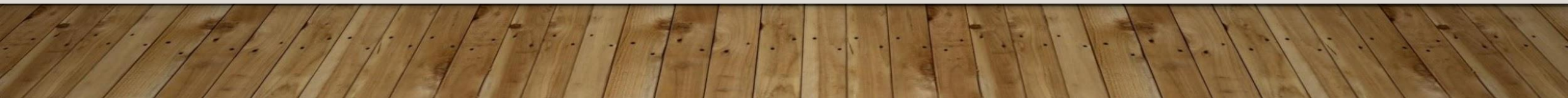


WHAT IS REFLEX?

Reflex is defined as the automatic rapid response to a stimulus.

REFLEX CONTROL OF MASTICATION

- Reflex control of mastication refers to involuntary rhythmic jaw movements during chewing initiated by sensory input (i.e. food in the mouth) that automatically activate muscles through brain stem reflexes.
- It involves automatic jaw opening & closing reflexes triggered by periodontal sensors controlling bite force.
- It is an adaptive type of control i.e. the body adjusts force based on the texture of food.
- E.g. gentle chewing while eating soft food & stronger chewing when the food is hard.



PURPOSE OF REFLEX CONTROL OF MASTICATION

- Protection (avoid injury)
 - Efficiency (chewing)
 - Co-ordination (smooth movement)
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JAW REFLEXES

Based on their function, jaw reflexes can be classified as:

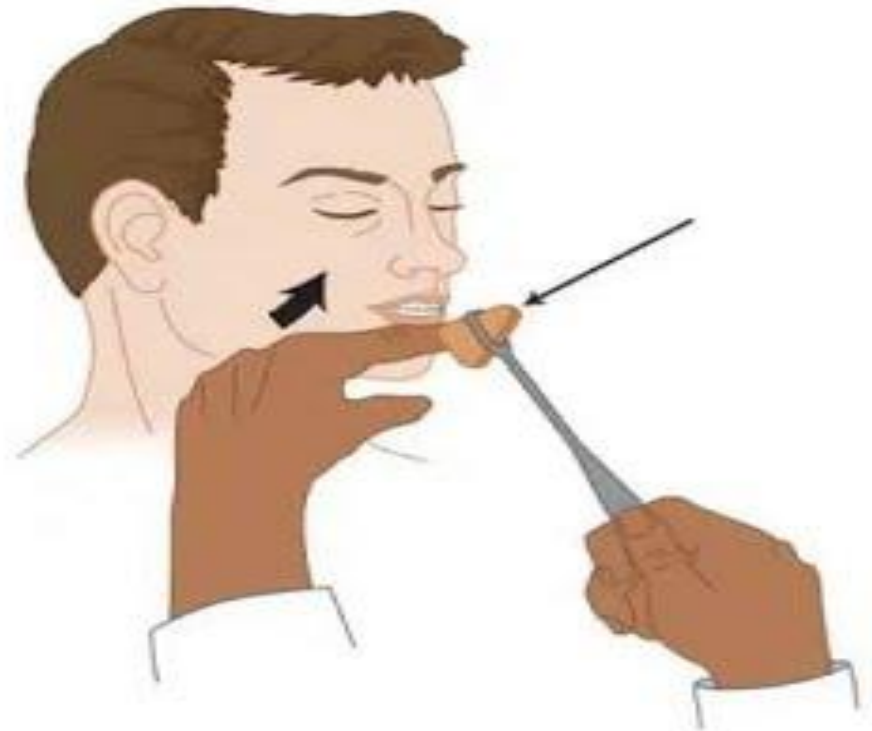
1. Stretch reflex -> maintains tone
2. Opening reflex -> protective function
3. Closing reflex -> maintains bite
4. Periodontal reflex -> fine control of forces during chewing.

I. JAW STRETCH REFLEX (JAW JERK)

- It is a monosynaptic response (very fast).
- Tested by tapping the chin.
- Exaggerated response shows upper motor neuron lesion.
- Stimulus: sudden stretch (e.g. tapping of chin)
- Receptors: muscle spindles of masseter muscle.
- Afferent pathway: trigeminal nerve
- Center: mesencephalic nucleus.
- Efferent pathway: motor nucleus.
- Response: quick contraction results in closing of the jaw.

JAW JERK/ REFLEX

- **Stimulus:** Tapping the middle of chin with slightly opened mouth
- **Response:** Closure of Mouth
- **Spinal Segment:** 5th Cranial Nerve

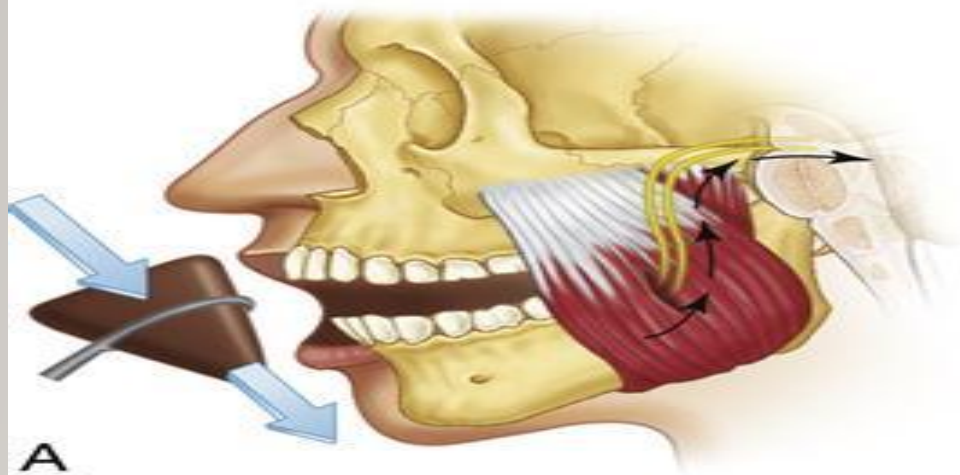


2. JAW OPENING REFLEX

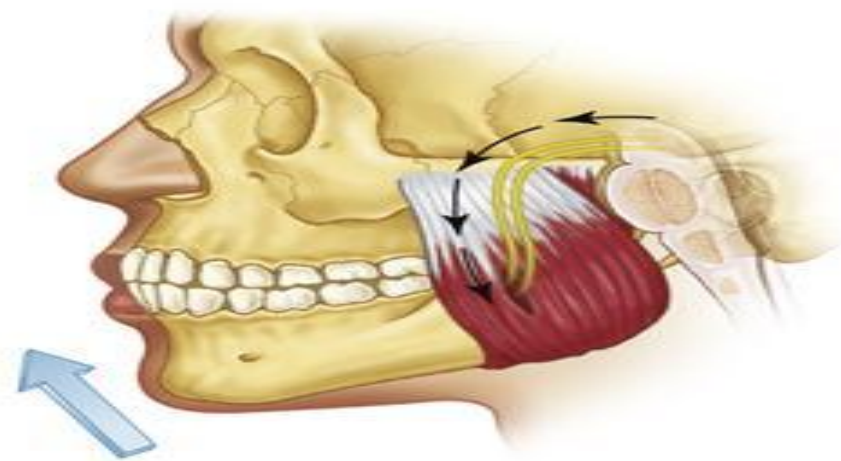
- It is an involuntary protective brain stem mechanism triggered by orofacial stimuli (e.g. pain from tongue, teeth, TMJ or oral mucosa) as well as mechanical stimulation.
- Causes immediate contraction of muscles that open the jaw (diagestric, lateral pterygoid)
- Inhibits jaw closing muscles.
- Function: prevent damage to the teeth & oral soft tissues.
- e.g. when we bite something too hard, jaw opens instantly.

3. JAW CLOSING REFLEX

- It is a monosynaptic stretch reflex also called ***masseter reflex***.
- This reflex maintains the force of closing phase of mastication ensuring efficient food crushing.
- This reflex is initiated when muscle spindles in the jaw closing muscles(masseter & temporalis) detect a stretch as they encounter food resistance.
- Receptors: periodontal ligament.
- Function: maintains bite force during chewing e.g. stronger closure of jaw when chewing hard food.



A



Trigeminal
mesencephalic nucleus

Trigeminal main
sensory nucleus

Spindle
afferent fiber

γ -Efferent
fiber

α -Efferent
fiber

Muscle
spindle

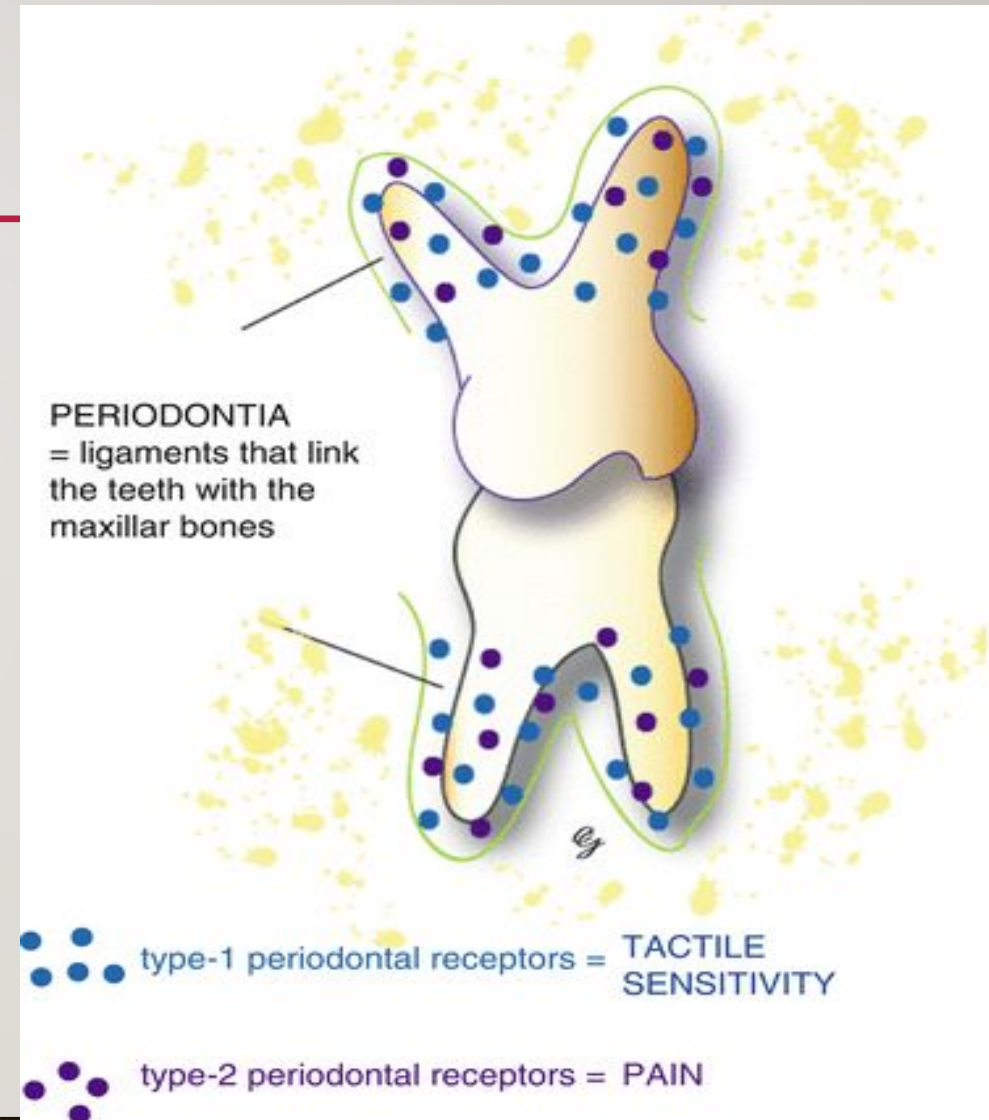
Jaw-closing
muscle

B

Trigeminal
motor nucleus

4. PERIODONTAL REFLEX

- Periodontal reflex is a brain stem mediated inhibitory reflex triggered by periodontal mechanoreceptors (PMRs) in the ligament.
- It regulates chewing force by causing a sudden reflex inhibition of jaw closing muscles (e.g. masseter) when high pressure is detected on a tooth.
- Function: adjust bite force to prevent tooth damage.
- E.g. periodontal reflex reduces the force of mastication to avoid tooth fracture when eating hard object.



ROLE OF TRIGEMINAL NERVE

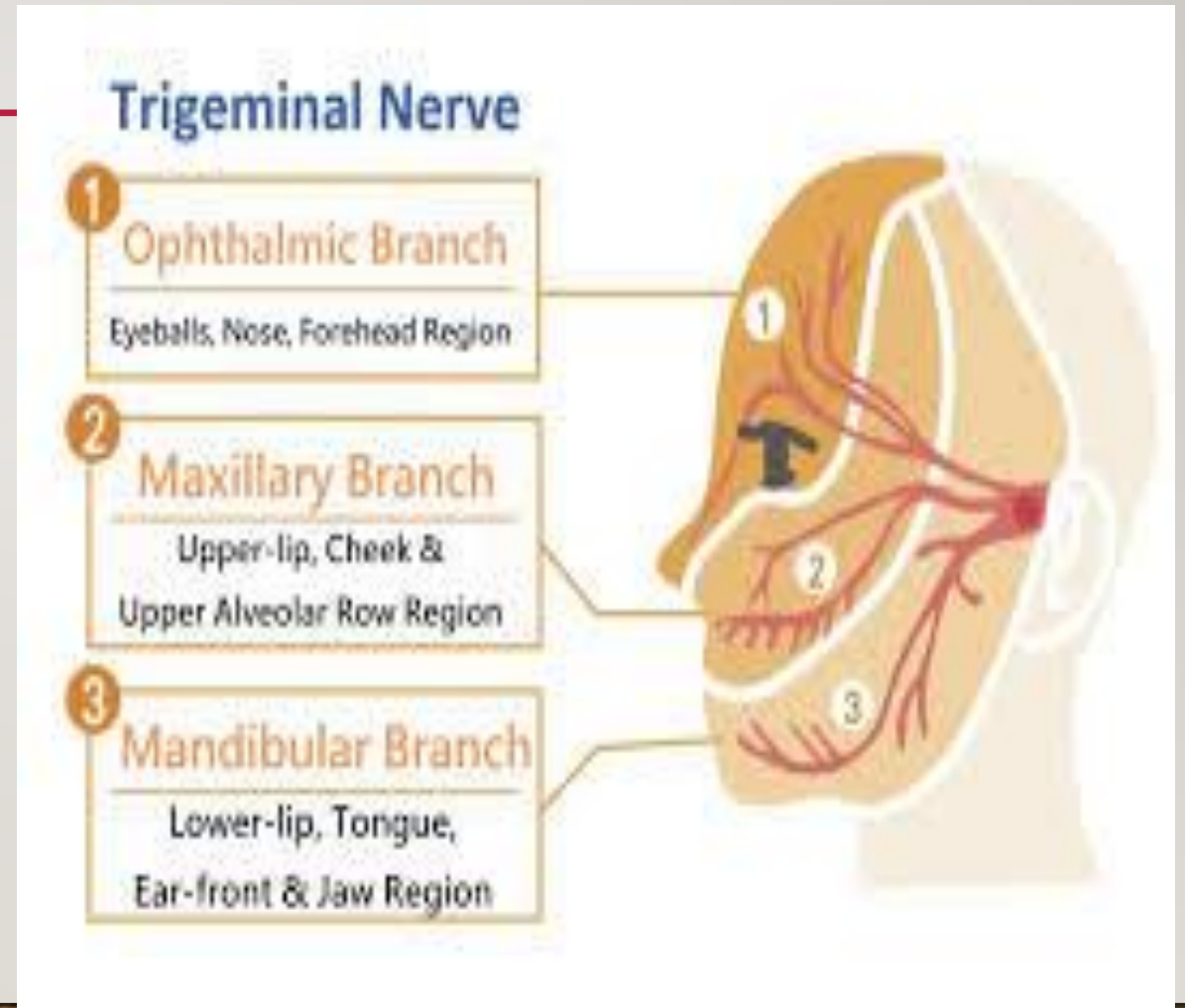
- Trigeminal nerve is the main nerve of mastication.
- It integrates sensory input & motor output.

Sensory function:

Teeth, gums & oral mucosa

Motor function:

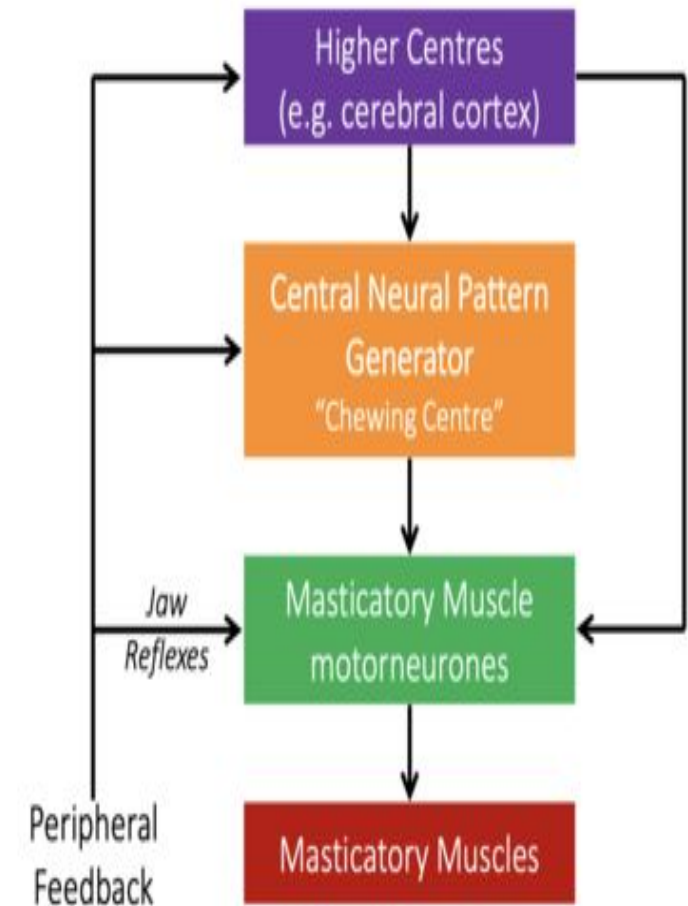
Muscles of mastication



CENTRAL CONTROL OF MASTICATION

- CPG creates the basic rhythm of opening & closing the jaw automatically without conscious effort.
- Central control of mastication is achieved by **central pattern generator (CPG)** located in the brain stem.
- The cerebral cortex acts as initiator & supervisor of process of mastication.
- The jaw opening reflex (activated by pressure) & jaw closing reflex (activated by muscle stretch) work within the CPG not only to protect the teeth but also ensure effective chewing.

Control of mastication



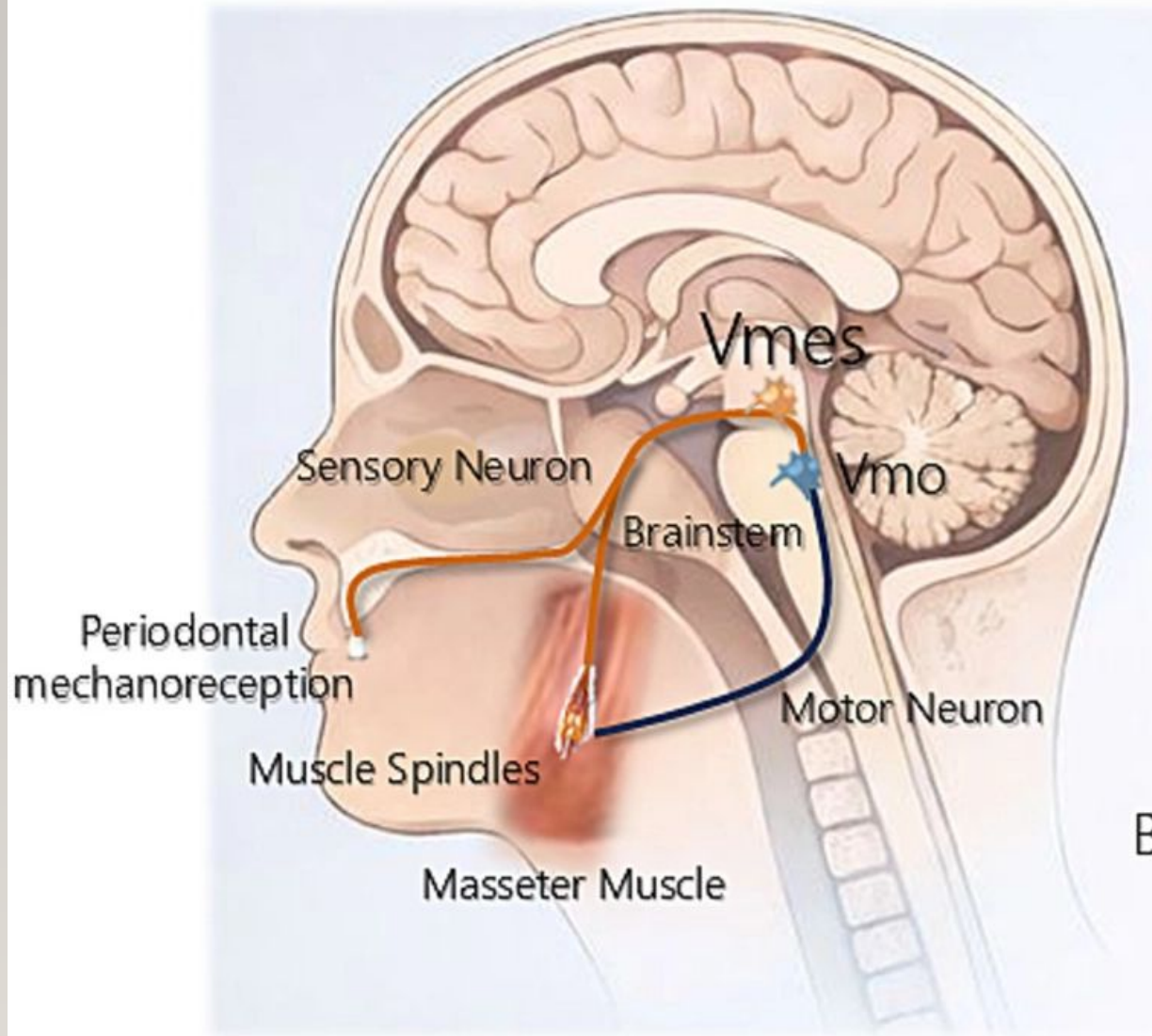
BRAINSTEM MASTICATORY CENTRAL PATTERN GENERATOR (CPG)

- The CPG is the neuronal network located in the brainstem (pons & medulla) that generates rhythmic motor commands for chewing even without voluntary input.
- It creates alternating opening & closing motions by driving motor neurons in the trigeminal motor nucleus, facial nucleus & hypoglossal nucleus.

RHYTHMOGENIC NEURONS

A group of neurons in the dorsal part of trigeminal sensory nucleus that has intrinsic bursting capabilities to produce the basic rhythm.





Jaw Jerk Reflex

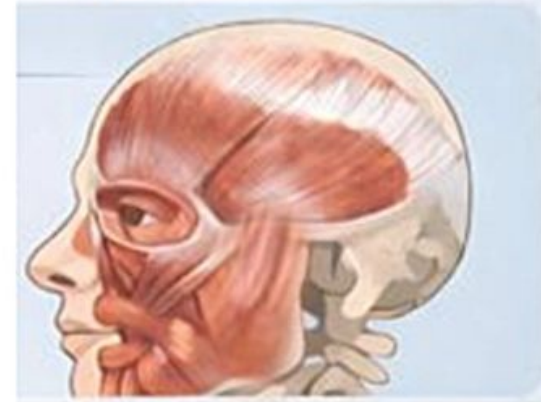
It is a monosynaptic reflex in which Vmes serves as the central relay of the afferent pathway.

Masticatory Movement

Vmes transmits proprioceptive inputs from jaw muscle spindles and periodontal receptors, contributing to the precise control of masticatory movements.

Bruxism, Temporomandibular Disorders

Vmes is involved in the exacerbation of pathology and the persistence of symptoms.



AFFERENT PATHWAY (SENSORY INPUT)

- Sensory input from the oral cavity modifies the CPG's activity allowing it to adapt to food texture & hardness.
- 1. ***Periodontal mechanoreceptors:*** sensory receptors in the tooth ligaments that detect biting force.
- 2. ***Muscle spindles:*** proprioceptors in the jaw muscles (especially closing muscles) that detect muscle stretch.
- 3. ***TMJ receptors:*** monitor the position of the joint.

EFFERENT PATHWAY (MOTOR COMMAND OUTPUT)

□ Motor commands travel from the brainstem to the muscles via specific nerves.

1. **Trigeminal nerve (CN V):** The mandibular branch (V_3) is the main nerve supplying the muscles of mastication.
2. **Facial nerve (CN VII):** It supplies the buccinator muscle of cheek to prevent food from escaping during chewing.
3. **Hypoglossal nerve (CN XII):** Controls muscles of the tongue.

MASTICATION DYSFUNCTION (CHEWING DIFFICULTY)

- ❑ Dysfunction of mastication involves impaired ability to bite, chew or swallow food often caused by TMJ disorders, missing teeth or muscle weakness.
- ❑ It affects about 11 to 12 million of US adults often causing jaw pain, restricted movement & nutritional risks.
- ❑ Treatment includes dental care, physical therapy or medication.

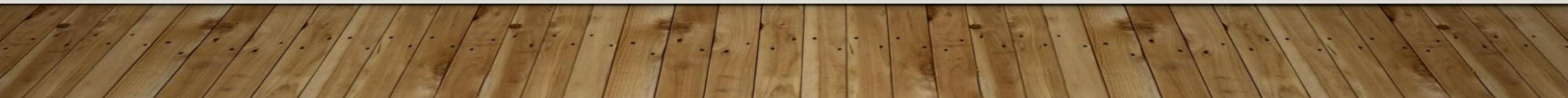


CAUSES OF CHEWING DIFFICULTY

1. **TMJ disorders:** dysfunction in the jaw joint/muscles causing clicking, pain or limited mouth opening.
 2. **Dental issues:** missing teeth, poorly fitting dentures or tooth loss significantly reduce bite force.
 3. **Muscle dysfunction:** reduced tongue strength & atrophy of muscles involved in mastication.
 4. **Other factors:** clenching/grinding of teeth(bruxism), trauma or arthritis.
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SYMPTOMS & ASSOCIATED ISSUES

1. **Pain & stiffness:** fatigue in jaw muscles often from chronic clenching.
2. **Reduced function:** difficulty opening the mouth or chewing efficiently which can impact nutrition in elderly people.
3. **Orofacial muscle issues:** associated with reduced strength of tongue muscles.



Causes



Jaw Misalignment
bite don't fit properly



TMJ Disorder
joint dysfunction



Bruxism
stresses the jaw joint



Injury/Trauma
affects movement



Arthritis
wears down the joint



Tight Jaw Muscles
tension restricts
smooth movement

MANAGEMENT & TREATMENT

1. ***Professional evaluation:*** assessment of TMJ & orofacial muscles function.
2. ***Dental intervention:*** addressing missing teeth & adjustment of occlusion.
3. ***Behavioral & physical therapy:*** exercises for strengthening or relaxing jaw muscles.
4. ***Oral appliances:*** occlusal plates/splints to prevent wear from grinding.

POST ASSESSMENT

Q1. The primary nerve responsible for motor control of mastication is:

- a) Facial nerve
- b) Glossopharyngeal nerve
- c) Trigeminal nerve
- d) Hypoglossal nerve

Q2. A light tap on the chin produces a brief contraction of jaw closing muscles. This response is best described as:

- a) Polysynaptic withdrawal reflex
- b) Conditioned reflex
- c) Monosynaptic stretch reflex
- d) Autonomic reflex

CONT..

Q3. The jaw opening reflex is primarily:

- a) A stretch reflex
- b) A protective reflex
- c) An autonomic reflex
- d) A voluntary reflex

Q4. During mastication sensory input from periodontal receptors mainly functions to:

- a) Initiate reflex swallowing
- b) Control tongue movements
- c) Produce rhythmic chewing
- d) Regulate bite force